# Experiment-3.3

**Aim:-** To Perform scanning while performing Malware Analysis

**Mapped CO: -** Understand the different types of Attacks, Vulnerabilities. **CO3**

## Objective: -

- To understand the role of scanning in malware analysis.
- To perform static and dynamic scanning of suspicious files using malware analysis tools.
- To identify malicious indicators such as signatures, behaviors, and anomalies.
- To correlate scan results with malware characteristics for preliminary classification.

## Theory: -

Malware analysis is the process of examining suspicious files or activities to determine whether they are malicious and to understand their behavior, origin, and impact. **Scanning** is a crucial initial step in malware analysis and is often categorized into:

### 1. Static Scanning

Static scanning involves analyzing a file without executing it. It includes:

- Antivirus signature-based scanning
- Hash analysis (MD5, SHA-1, SHA-256)
- String analysis
- File header and metadata inspection

Static scanning is safe and fast but may fail to detect new or obfuscated malware.

### 2. Dynamic Scanning

Dynamic scanning involves executing the suspicious file in a controlled environment (sandbox or virtual machine) to observe its behavior. It includes:

- Process monitoring
- Network traffic analysis
- File system and registry changes

Dynamic scanning helps in detecting zero-day and polymorphic malware but requires careful isolation to avoid system infection.

### Importance of Scanning in Malware Analysis

- Provides early detection of known malware
- Helps prioritize samples for deeper analysis
- Reduces analysis time by filtering benign files
- Supports incident response and forensic investigations

## Tools Required: -

- Isolated Virtual Machine (Windows/Linux)
- Antivirus Scanner (e.g., Windows Defender, ClamAV)
- Malware Scanning Platforms (e.g., VirusTotal – for hash analysis only)
- Process Monitor / Process Explorer
- Network Monitoring Tool (e.g., Wireshark)
- Sample suspicious file (test malware or EICAR test file)

## Steps / Procedure: -

### Step 1: Access VirusTotal

1. Open a web browser.
2. Go to **https://www.virustotal.com**
3. Sign in using a Google or VirusTotal account (recommended for full features).

### Step 2: Choose the Scanning Method

VirusTotal allows **three main scanning options**:

- **File Scan**
- **URL Scan**
- **Hash Scan**

Select the appropriate option based on the sample type.

### Step 3: Upload the Suspicious File (File Scanning)

1. Click on the **"File"** tab.
2. Click **"Choose file"** or drag and drop the suspicious file.
3. Ensure the file size is within VirusTotal's limit.
4. Click **"Confirm upload"** to begin scanning.

- VirusTotal will scan the file using **70+ antivirus engines**.

### Step 4: Scan Using File Hash (Hash-Based Scanning)

1. Click on the **Search** bar.
2. Paste the **file hash** (MD5 / SHA-1 / SHA-256).
3. Press **Enter**.
4. If the file was previously analyzed, VirusTotal displays existing results.

- This method avoids uploading malicious files again.

### Step 5: Scan Suspicious URLs

1. Click on the **"URL"** tab.
2. Enter the suspicious website link.
3. Click **"Search"** or **"Scan URL"**.
4. VirusTotal checks the URL against multiple security databases.

### Step 6: Analyze Detection Results

1. Review the **detection ratio** (e.g., `12/70`).
2. Identify malware classifications such as:
   - Trojan
   - Ransomware
   - Spyware
   - Worm
3. Check **AV engine names** that detected the malware.

### Step 7: Examine Detailed Analysis Tabs

Navigate through the analysis sections:

- **Details** – File type, size, hashes, signatures
- **Behavior** – Suspicious runtime activities
- **Relations** – Connected IPs, domains, and files
- **Community** – User comments and insights

### Step 8: Interpret Behavioral Indicators

Look for red flags such as:

- Registry modifications
- Suspicious API calls
- Network connections to malicious IPs
- Persistence mechanisms

### Step 9: Generate and Save the Report

1. Click on **"Download"** or **"Share"**.
2. Save the report for documentation or further analysis.

3. Use report data for incident response or academic submission.

### Step 10: Conclude the Analysis

1. Decide whether the file/URL is **Malicious, Suspicious, or Safe**.
2. Recommend appropriate actions:
    - Quarantine
    - Removal
    - Network blocking
    - Further sandbox analysis

## Result: -

The suspicious file was successfully scanned using static and dynamic malware analysis techniques. The scanning process helped in identifying malicious indicators such as known signatures, abnormal process behavior, and suspicious network activity. This experiment demonstrated the importance of scanning as a preliminary and essential step in malware analysis.

## YouTube Video Link: -

1. https://www.youtube.com/watch?v=KNe4hTVhpPQ
2. https://www.youtube.com/watch?v=i2I37T23mpI
3. https://www.youtube.com/watch?v=uHhKkLwT4Mk&list=PLBf0hzazHTGMSlOI2HZGc08ePwut6A2Io

## Viva Questions

1. What is malware analysis?
2. Why is scanning important in malware analysis?
3. Differentiate between static and dynamic scanning.
4. What are malware signatures?
5. What is hash-based malware detection?
6. Why should malware analysis be performed in a virtual machine?
7. What risks are involved in dynamic malware analysis?
8. Name any tools used for malware scanning.
9. What is the EICAR test file?
10. How does behavioural analysis help in detecting zero-day malware?